Networking Guide

From Hubs & Switches to VLANs and ACLs in Cisco Packet Tracer

Networking Fundamentals throughout Practical Labs

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Part I

Foundations — Devices, LANs & Subnetting

Chapter 1

Introduction to Networking Devices

1.1 Why Networking Devices Matter

Every network — from a two-PC home setup to an enterprise campus — is built from three fundamental "connecting" devices: the **Hub**, the **Switch**, and the **Router**. Understanding exactly what each device does at which OSI layer is the single most important foundation for everything that follows (VLANs, routing, ACLs).

1.2 The Hub (Layer 1 Device)

A hub is a dumb, multi-port repeater. It operates purely at **Layer 1 (Physical)**.

- Receives an electrical signal on one port and **blindly repeats it out of every other port**.
- Has **no MAC address table** — it does not understand frames or addresses.
- All ports share **one single collision domain** and **one broadcast domain**.
- Operates in **half-duplex** only → causes collisions as more devices are added.

WARNING

Hubs are obsolete in real networks and in Packet Tracer are mainly used to *demonstrate* collision domains. Never use a hub in a design task unless specifically asked to illustrate collisions.

1.3 The Switch (Layer 2 Device)

A switch is an intelligent, multi-port bridge operating at **Layer 2 (Data Link)**.

- Learns source **MAC addresses** of incoming frames and builds a **MAC Address Table (CAM table)**.
- Forwards frames *only* to the port where the destination MAC resides (unicast) — unknown/broadcast/multicast frames are flooded.
- **Each switch port is its own collision domain** (because of full-duplex, point-to-point links).
- **All ports on a switch remain in one broadcast domain** — unless VLANs are configured (Part II).

TIP

In Packet Tracer, click a switch while frames are moving (Simulation Mode) and open its MAC table under `CLI > show mac address-table` to visually understand learning.

1.4 The Router (Layer 3 Device)

A router works at **Layer 3 (Network)**, making forwarding decisions based on **IP addresses**.

- Each router interface belongs to a **different network/subnet** and is a **separate broadcast domain**.
- Maintains a **Routing Table** (built via connected routes, static routes, or dynamic routing protocols).
- Used to connect multiple LANs/VLANs together, and connect LANs to the Internet/WAN.

Device	OSI Layer	Collision Domain	Broadcast Domain
Hub	1 (Physical)	1 per hub	1 per hub
Switch	2 (Data Link)	1 per port	1 per switch (no VLAN)
Router	3 (Network)	1 per port	1 per port/interface

Chapter 2

Building LAN in Packet Tracer

2.1 What Is a LAN?

A **Local Area Network (LAN)** is a group of devices (PCs, printers, servers) connected together, typically within one building/location, sharing the **same subnet/broadcast domain**, usually via a switch.

2.2 LAN Using a Switch Only

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Topology: 3 PCs → connected to **Switch2960** using straight-through (copper) cables.
Steps in Packet Tracer:

1. Drag 1 Switch (2960) and 3 PCs onto the workspace.
2. Connect PC0, PC1, PC2 to switch FastEthernet0/1, 0/2, 0/3 using **Automatic** or **Copper Straight-through** cable.
3. Click each PC → Desktop → IP Configuration → assign:

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0

TIP

Test connectivity using Command Prompt on any PC:

```
C:\> ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

2.3 LAN Using a Router (Connecting Two Separate LANs)

A single switch LAN cannot talk to another LAN on a different subnet without a router.

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Topology:

- LAN1: PC0, PC1 → Switch0 → Router G0/0/0
- LAN2: PC2, PC3 → Switch1 → Router G0/0/1

CHEAT SHEET: Basic Router Interface Configuration

```
Router> enable
Router# configure terminal
Router(config)# interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# exit
Router# write memory      ! saves running-config to startup-config
```

NOTE

Each PC must use the router's connected interface IP as its **Default Gateway**:

- PC0, PC1 Gateway = 192.168.1.1
- PC2, PC3 Gateway = 192.168.2.1

Without this gateway, inter-LAN communication will fail even if IPs are correctly set.

2.4 Verifying the Router

```
Router# show ip interface brief
Router# show running-config
Router# show ip route
```

Chapter 3

Subnetting & Subnet Masking

3.1 IP Addressing Recap

An IPv4 address is 32 bits, written as 4 octets (e.g., `192.168.1.10`). It is divided into:

- **Network portion** — identifies the LAN/subnet.
- **Host portion** — identifies the device within that subnet.

3.2 The Subnet Mask

The subnet mask tells a device which bits are "network" and which are "host" bits.

CIDR	Subnet Mask	Hosts Usable
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/29	255.255.255.248	6
/30	255.255.255.252	2 (point-to-point link)

NOTE

Formula for usable hosts: $2^h - 2$, where h = number of host bits (the -2 removes the Network ID and Broadcast ID).

3.3 CIDR Notation

CIDR (**Classless Inter-Domain Routing**) notation expresses the mask as a slash + number of network bits, e.g. `192.168.1.0/24` = 24 network bits, 8 host bits.

3.4 Step-by-Step Subnetting Example

Problem: Subnet `192.168.10.0/24` into 4 equal subnets.

1. Number of subnets needed = 4 ⇒ borrow bits so $2^n \geq 4 \Rightarrow n = 2$ bits borrowed.
2. New mask = /24 + 2 = /**26** = 255.255.255.192

3. Block size = 256 − 192 = 64

Subnet	Network ID	Usable Range	Broadcast
1	192.168.10.0	.1 – .62	192.168.10.63
2	192.168.10.64	.65 – .126	192.168.10.127
3	192.168.10.128	.129 – .190	192.168.10.191
4	192.168.10.192	.193 – .254	192.168.10.255

TIP

Shortcut trick ("Magic Number" method): Block size = 256 − interesting octet value of mask. Keep adding the block size to get each subnet's Network ID.

3.5 VLSM (Variable Length Subnet Masking)

VLSM allows subnets of **different sizes** from the same base network — essential for efficient router/VLAN designs where one LAN needs 50 hosts and another only needs 2 (e.g., a router-to-router link).

CHEAT SHEET: Subnetting Quick Reference

```
/24 = 255.255.255.0     -> 256 addresses / 254 hosts
/25 = 255.255.255.128   -> 128 addresses / 126 hosts
/26 = 255.255.255.192   -> 64 addresses  / 62 hosts
/27 = 255.255.255.224   -> 32 addresses  / 30 hosts
/28 = 255.255.255.240   -> 16 addresses  / 14 hosts
/29 = 255.255.255.248   -> 8  addresses  / 6 hosts
/30 = 255.255.255.252   -> 4  addresses  / 2 hosts  (router links)
/32 = 255.255.255.255   -> 1 address (host route)
```

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In Packet Tracer, build the 4-subnet design above:

1. Create 4 separate switch LANs with 2 PCs each.
2. Assign IPs per the subnet table above.
3. Connect each LAN to a dedicated router interface with the matching **Network ID's first usable IP** as gateway.
4. Verify using `ping` across LANs after setting gateways correctly.

Part II

VLANs & Access Control Lists

Chapter 4

Understanding VLANs

4.1 What Is a VLAN?

A **Virtual LAN (VLAN)** logically segments a single physical switch into multiple separate broadcast domains — *without* needing separate physical switches.

- Each VLAN = its own broadcast domain = its own subnet.
- Devices in different VLANs **cannot** communicate without a Layer-3 device (router or Layer-3 switch).
- Benefits: security, reduced broadcast traffic, logical grouping (e.g., Sales VLAN, HR VLAN) regardless of physical location.

4.2 Access Ports vs Trunk Ports

- **Access Port:** Belongs to exactly one VLAN; connects to end devices (PCs, printers).
- **Trunk Port:** Carries traffic for **multiple VLANs** between switches or between switch and router, using **802.1Q tagging**.

Port Type	Used Between
Access	Switch ↔ End device
Trunk	Switch ↔ Switch, Switch ↔ Router

Chapter 5

Router-on-a-Stick

5.1 Topology

- 1 Router (single physical interface, e.g., G0/0)
- 1 Switch (2960)
- VLAN 10 (Sales) — PC0, PC1
- VLAN 20 (HR) — PC2, PC3
- Switch trunk port connects to Router G0/0 (the router interface is divided into **sub-interfaces**, hence "router-on-a-stick")

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Physical Cabling:

- PC0, PC1 → Switch Fa0/1, Fa0/2 (VLAN 10)
- PC2, PC3 → Switch Fa0/3, Fa0/4 (VLAN 20)
- Switch Gig0/1 → Router G0/0 (Trunk link)

5.2 Step 1: Create VLANs on the Switch

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
```

5.3 Step 2: Assign Access Ports to VLANs

```
Switch(config)# interface range fastEthernet0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit

Switch(config)# interface range fastEthernet0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

5.4 Step 3: Configure Trunk Port Toward Router

```
Switch(config)# interface gigabitEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# exit
```

5.5 Step 4: Configure Router Sub-Interfaces

```
Router> enable
Router# configure terminal

Router(config)# interface gigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface gigabitEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

Router(config)# interface gigabitEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

NOTE

`encapsulation dot1Q 10` must match the VLAN ID exactly. Each sub-interface acts as the **default gateway** for its VLAN.

5.6 Step 5: PC Configuration

PC	VLAN	IP	Gateway
PC0	10	192.168.10.2	192.168.10.1
PC1	10	192.168.10.3	192.168.10.1
PC2	20	192.168.20.2	192.168.20.1
PC3	20	192.168.20.3	192.168.20.1

5.7 Verification Commands

```
Switch# show vlan brief
Switch# show interfaces trunk
Router# show ip interface brief
Router# show running-config
```

Chapter 6

Switch Inter-VLAN Routing

6.1 Concept: Layer 3 Switch with SVIs

Without a router, a **Layer 3 (Multilayer) Switch** (e.g., Cisco 3560/3650 in Packet Tracer) can route between VLANs internally using **Switch Virtual Interfaces (SVIs)**.

NOTE

This method requires a **Multilayer Switch** — a normal Layer 2 switch (2960) **cannot** route between VLANs on its own.

6.2 Topology

- 1 Multilayer Switch (e.g., 3560)
- VLAN 10 — PC0, PC1 on Fa0/1, Fa0/2
- VLAN 20 — PC2, PC3 on Fa0/3, Fa0/4
- No external router needed

6.3 Step 1: Create VLANs & Assign Ports

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)# exit

Switch(config)# interface range fastEthernet0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Switch(config)# interface range fastEthernet0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

6.4 Step 2: Enable Routing & Create SVIs

```
Switch(config)# ip routing
! Critical step - turns on Layer 3 routing on the switch

Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit

Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
```

WARNING

Forgetting `ip routing` is the #1 mistake — without it, SVIs will have "up/up" status but will **not route** between VLANs.

6.5 Step 3: PC Gateway Configuration

PC	VLAN	IP	Gateway
PC0	10	192.168.10.2	192.168.10.1
PC1	10	192.168.10.3	192.168.10.1
PC2	20	192.168.20.2	192.168.20.1
PC3	20	192.168.20.3	192.168.20.1

6.6 Verification

```
Switch# show ip route
Switch# show vlan brief
Switch# show ip interface brief
```

CHEAT SHEET: VLAN Commands Summary

```
vlan <id>                          ! create VLAN
name <name>                        ! name the VLAN
switchport mode access             ! set port as access
switchport access vlan <id>        ! assign VLAN to port
switchport mode trunk              ! set port as trunk
switchport trunk allowed vlan <list>
interface vlan <id>                ! create SVI (L3 switch)
ip routing                         ! enable L3 switch routing
encapsulation dot1Q <id>           ! router sub-interface tagging
show vlan brief
show interfaces trunk
show ip interface brief
```

Chapter 7

Access Control Lists (ACLs)

7.1 What Is an ACL?

An **Access Control List** is a set of rules applied to a router (or L3 switch) interface to **permit or deny** traffic based on defined criteria. Processed **top-down**, first match wins, and there is an **implicit deny all** at the end of every ACL.

7.2 Standard ACL

7.2.1 Characteristics

- Number range: **1–99** and **1300–1999** (expanded range)
- Filters based **only on Source IP Address**
- Best practice: place **close to the destination** (since it can't filter by destination/port, placing it near source would block too much).

7.2.2 Syntax

```
Router(config)# access-list <1-99> {permit|deny} <source-ip> <wildcard-mask>
```

7.2.3 Example: Block PC on 192.168.10.5 from reaching network 192.168.20.0/24

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Topology: Two LANs connected via router as in Chapter 2. Goal: deny host 192.168.10.5, allow everyone else.

```
Router> enable
Router# configure terminal
Router(config)# access-list 10 deny host 192.168.10.5
Router(config)# access-list 10 permit any
Router(config)# exit
```

7.2.4 Applying to an Interface

```
Router(config)# interface gigabitEthernet0/1
Router(config-if)# ip access-group 10 out
Router(config-if)# exit
```

NOTE

Wildcard masks are the inverse of subnet masks. E.g., for network 192.168.20.0/24, wildcard = 0.0.0.255.
Shortcut: `host 192.168.10.5` = `192.168.10.5 0.0.0.0`.
`any` = `0.0.0.0 255.255.255.255`

7.3 Extended ACL

7.3.1 Characteristics

- Number range: **100–199** and **2000–2699** (expanded range)
- Filters based on **Source IP, Destination IP, Protocol (TCP/UDP/ICMP), and Port Numbers**
- Best practice: place **close to the source** (filters precisely, so blocking early saves bandwidth).

7.3.2 Syntax

```
Router(config)# access-list <100-199> {permit|deny} <protocol> \
   <source-ip> <wildcard> <destination-ip> <wildcard> [eq <port>]
```

7.3.3 Example: Block only HTTP (Web) from LAN1 to Server, allow everything else

```
Router(config)# access-list 110 deny tcp 192.168.10.0 0.0.0.255 \
   host 192.168.20.100 eq 80
Router(config)# access-list 110 permit ip any any
```

7.3.4 Applying to an Interface

```
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip access-group 110 in
Router(config-if)# exit
```

WARNING

Always end every custom extended ACL with `permit ip any any` (if required) — otherwise the **implicit deny** will silently block ALL other traffic.

7.4 Named ACLs (Alternative, Readable Syntax)

```
Router(config)# ip access-list extended BLOCK_HTTP
Router(config-ext-nacl)# deny tcp 192.168.10.0 0.0.0.255 host 192.168.20.100
    eq 80
Router(config-ext-nacl)# permit ip any any
Router(config-ext-nacl)# exit

Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip access-group BLOCK_HTTP in
```

7.5 Verifying ACLs

```
Router# show access-lists
Router# show ip interface gigabitEthernet0/0
Router# show running-config
```

CHEAT SHEET: ACL Quick Reference

```
STANDARD ACL
  Range: 1-99, 1300-1999
  Filters by: Source IP only
  Place: Close to DESTINATION
  Syntax: access-list <num> {permit|deny} <src-ip> <wildcard>

EXTENDED ACL
  Range: 100-199, 2000-2699
  Filters by: Source IP, Dest IP, Protocol, Port
  Place: Close to SOURCE
  Syntax: access-list <num> {permit|deny} <proto> <src> <wc> <dst> <wc>
       [eq port]

APPLY TO INTERFACE
  ip access-group <num/name> {in|out}

COMMON WILDCARD MASKS
  /24 -> 0.0.0.255      /30 -> 0.0.0.3
  /25 -> 0.0.0.127      host -> 0.0.0.0
  /26 -> 0.0.0.63       any  -> 255.255.255.255 (addr 0.0.0.0)

VERIFY
  show access-lists
  show ip interface <intf>
```

Chapter 8

Cheat Sheet & Troubleshooting

8.1 Full Command Reference

CHEAT SHEET: Global Commands

```
enable                              ! privileged mode
configure terminal                  ! global config mode
hostname <name>                     ! rename device
exit / end                          ! move up a level
write memory / copy run start       ! save config
show running-config                 ! view active config
show startup-config                 ! view saved config
erase startup-config                ! factory reset (then reload)
reload                              ! restart device
```

CHEAT SHEET: Interface & IP Commands

```
interface <type><number>
ip address <ip> <mask>
no shutdown
description <text>
show ip interface brief
show interfaces
```

CHEAT SHEET: Routing

```
ip route <dest-network> <mask> <next-hop-ip>
show ip route
```

8.2 Common Troubleshooting Checklist

1. **Ping fails** → check IP/mask/gateway on PC via `ipconfig`.
2. **Interface down** → check `no shutdown` was issued; check cable type.
3. **VLAN traffic not passing** → confirm trunk allows the VLAN (`show interfaces trunk`).
4. **Inter-VLAN not working (L3 switch)** → confirm `ip routing` is enabled.
5. **ACL blocking everything** → check for missing `permit any` / implicit deny.

6. **Router-on-a-stick not routing** → verify `encapsulation dot1Q` matches VLAN ID on each sub-interface.

TIP

Use **Simulation Mode** in Packet Tracer and send a PDU between devices to watch exactly where a packet is dropped — this instantly reveals misconfigurations in trunking, routing, or ACLs.